IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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IBM Data Science Professional Certificate · Applied Data Science Capstone



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

How the analysis was done

- Collected 90 Falcon 9 launches (2010–2020) from the SpaceX REST API and Wikipedia
- Cleaned the data and created a landing label: Class 1 = landed, 0 = did not
- Explored patterns with SQL queries and seaborn charts
- Mapped launch sites in Folium; built an interactive Plotly Dash dashboard
- Tuned 4 classifiers with GridSearchCV (10-fold cross-validation)

What we found

- Landing success grew from 0% (2010–13) to 90% (2019)
- KSC LC-39A has the best success rate: 10 of 13 launches landed
- SSO and VLEO orbits land reliably; GTO is the hardest (52%)
- Every model scored 83.3% on the held-out test launches
- The Decision Tree had the best cross-validation score

90

Falcon 9 launches analyzed

66.7%

first stages landed overall

83.3%

model accuracy on test set

$100M+

price gap reuse can create

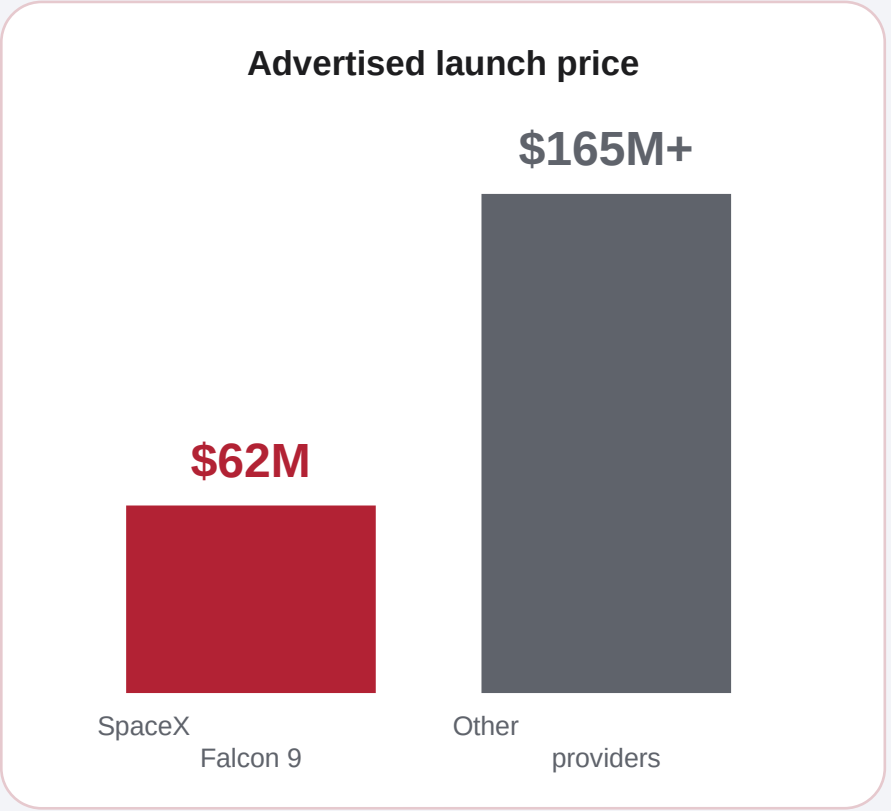
Introduction

Project background and context

- SpaceX advertises Falcon 9 launches at about $62 million. Other providers charge $165 million or more.
- Most of the savings comes from landing and reusing the first stage.
- If we can predict whether the first stage will land, we can estimate the real cost of a launch. That gives a new competitor, Space Y, a basis for bidding against SpaceX.

Questions this project answers

- Which factors (payload, orbit, launch site, booster, experience) drive a successful landing?
- How have landing outcomes changed over time and across sites?
- Can a machine learning model predict landing success, and how accurately?



Section 1

Methodology



Methodology

1

Data collection

SpaceX REST API (launch, rocket, launchpad, payload and core endpoints) plus web scraping of the Wikipedia Falcon 9 launch table

2

Data wrangling

Filtered to Falcon 9, filled missing payload mass with the mean, and turned 8 landing outcome types into a binary Class label

3

Exploratory data analysis

SQL queries on the launch table (SQLite) and seaborn/matplotlib charts; one-hot encoded features for modeling

4

Interactive visual analytics

Folium maps of sites, outcomes and proximities; Plotly Dash dashboard with a site dropdown and payload slider

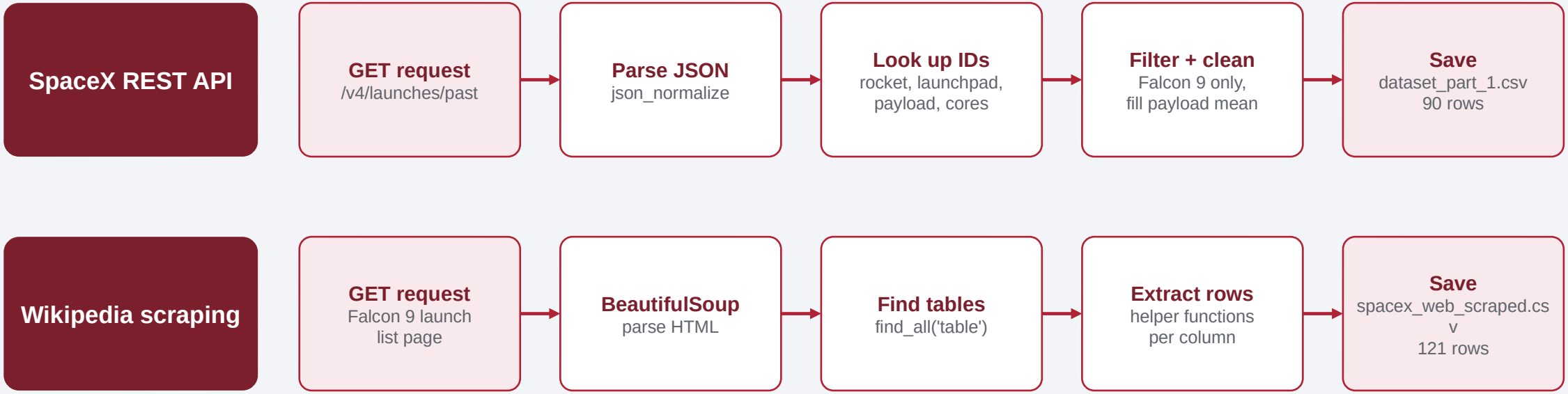
5

Predictive analysis

Standardized features, 80/20 split, GridSearchCV (cv=10) for Logistic Regression, SVM, Decision Tree and KNN; compared test accuracy and confusion matrices

Data Collection

Two sources were combined to describe every Falcon 9 launch: SpaceX's own REST API (structured JSON) and the Wikipedia launch table (HTML). Both end as a pandas DataFrame saved to CSV.

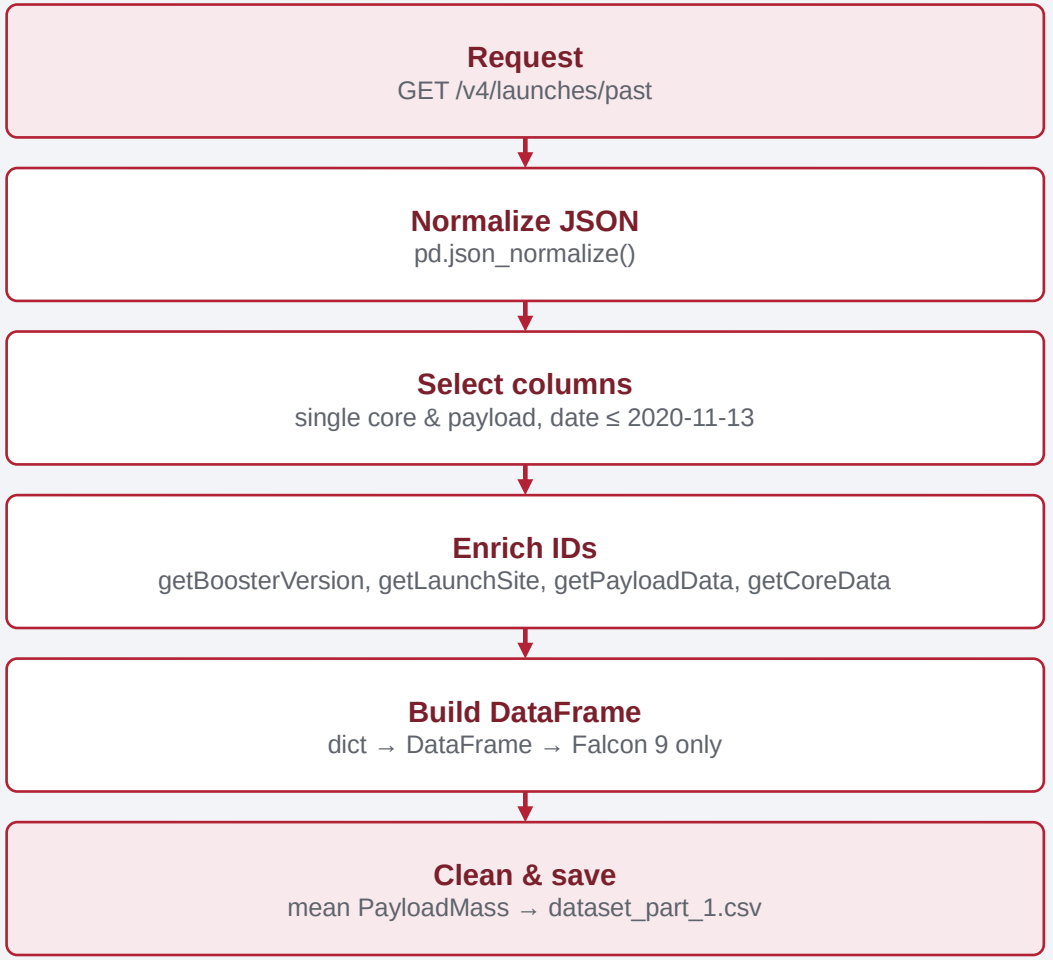


Takeaway When the live API returned errors (JSONDecodeError), I used IBM's saved copy of the same API response, so the pipeline kept running.

Data Collection – SpaceX API

Key phrases

- requests.get() on api.spacexdata.com/v4/launches/past
- .json() and pd.json_normalize() to flatten the response
- Kept rocket, payloads, launchpad, cores, flight number and date; launches on or before 13 Nov 2020
- Helper functions call the API again to turn IDs into booster version, launch site and coordinates, payload mass, orbit, and landing details
- Built a dictionary, then a DataFrame; kept Falcon 9 only (90 launches)
- Replaced missing PayloadMass with the column mean

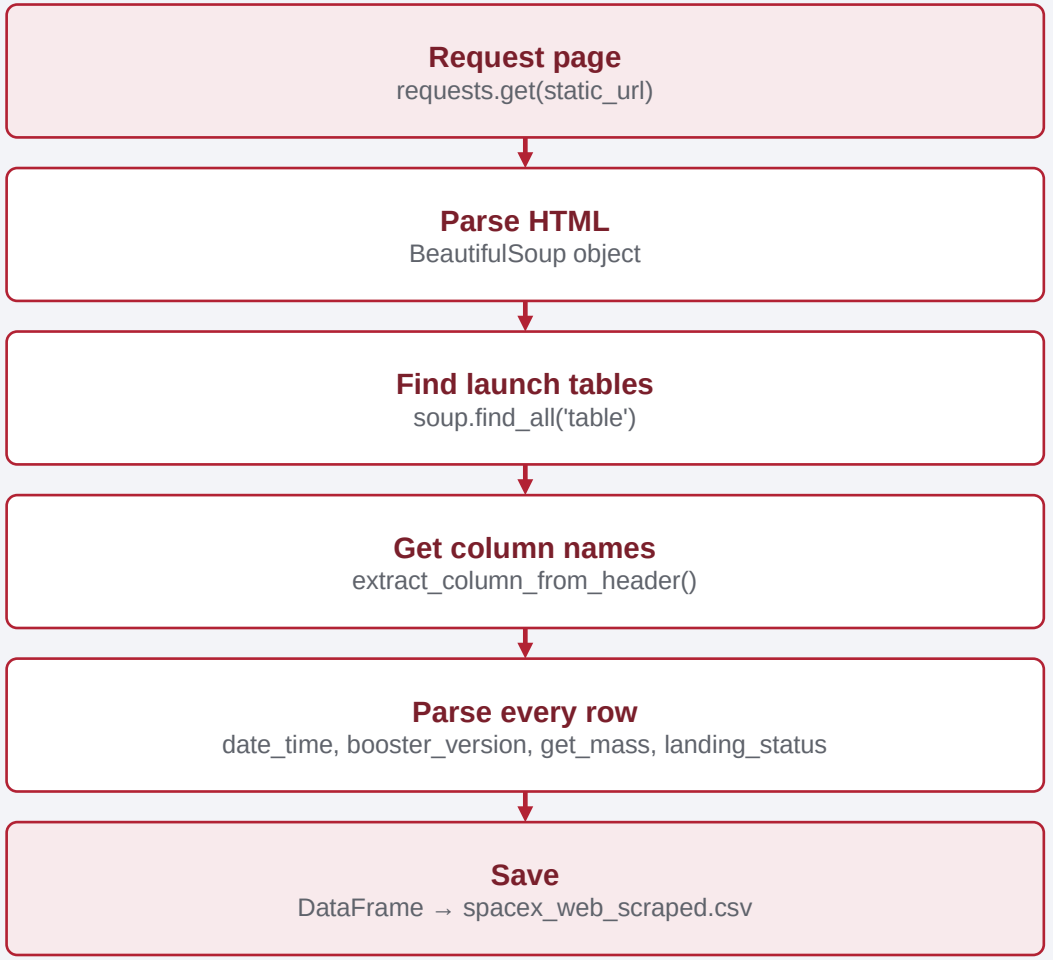


GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

Key phrases

- requests.get() on the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page (static snapshot)
- BeautifulSoup(response.text, 'html.parser')
- find_all('table') to get the launch tables
- Read column names from the <th> header cells
- Helper functions parse each row: date and time, booster version, payload mass, landing status
- Dictionary of lists → DataFrame → CSV



GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

Goal: turn 8 different landing outcome descriptions into one training label, Class (1 = landed, 0 = did not land).



Outcome	Meaning	Count	Class
`True ASDS`	Landed on drone ship	41	1
`None None`	No landing attempt	19	0
`True RTLS`	Landed on ground pad	14	1
`False ASDS`	Failed on drone ship	6	0
`True Ocean`	Controlled ocean landing	5	1
`False Ocean`	Failed ocean landing	2	0
`None ASDS`	No attempt (drone ship)	2	0
`False RTLS`	Failed on ground pad	1	0

Results

- 60 of 90 launches landed (66.7%)
- Launch sites: CCAFS SLC 40 (55), KSC LC 39A (22), VAFB SLC 4E (13)
- GTO is the most common orbit (27 launches)

GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/labs-jupyter-spacex-Data_wrangling-v2.ipynb

EDA with Data Visualization

Chart	Type	Why I used it
Flight Number vs. Launch Site	Scatter	See whether experience at each site improves landings
Payload Mass vs. Launch Site	Scatter	Check if some sites handle heavier payloads, and how that affects landing
Success Rate by Orbit	Bar	Compare landing success across orbit types
Flight Number vs. Orbit	Scatter	See whether success in each orbit depends on experience
Payload Mass vs. Orbit	Scatter	See how payload weight relates to landing for each orbit
Yearly Success Trend	Line	Show how landing success changed over time

Feature engineering

- Selected FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount and Serial
- One-hot encoded Orbit, LaunchSite, LandingPad and Serial with get_dummies(), then cast to float64: 83 numeric features for the models

GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

1. Unique launch site names (DISTINCT)
2. 5 records where the launch site begins with 'CCA' (LIKE, LIMIT)
3. Total payload mass carried for NASA (CRS) (SUM)
4. Average payload mass for booster version F9 v1.1 (AVG)
5. Date of the first successful ground-pad landing (MIN)
6. Boosters with a drone-ship success and payload between 4,000 and 6,000 kg (BETWEEN)
7. Total successful and failed mission outcomes (COUNT, GROUP BY)
8. Boosters that carried the maximum payload (subquery with MAX)
9. 2015 drone-ship landing failures by month, booster and site (substr)
10. Ranking of landing outcome counts, 2010-06-04 to 2017-03-20 (GROUP BY, ORDER BY DESC)

Loaded Spacex.csv into SQLite (my_data1.db) and queried it with the %sql magic in Jupyter.

GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map object	What it shows	Why I added it
folium.Circle + Marker (DivIcon)	One highlighted circle and text label per launch site	Shows exactly where the 4 sites are and how they cluster
MarkerCluster	One marker per launch, grouped by site	Keeps many launches at the same coordinates readable
Green / red Icon markers	Green = landed (class 1), red = failed (class 0)	Shows each site's success rate at a glance
MousePosition	Shows latitude and longitude under the cursor	Used to find the coordinates of a nearby coastline, railway, highway and city
PolyLine + distance Marker	Line from a site to each landmark, labelled in km	Measures how close sites are to transport, coast and cities

GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

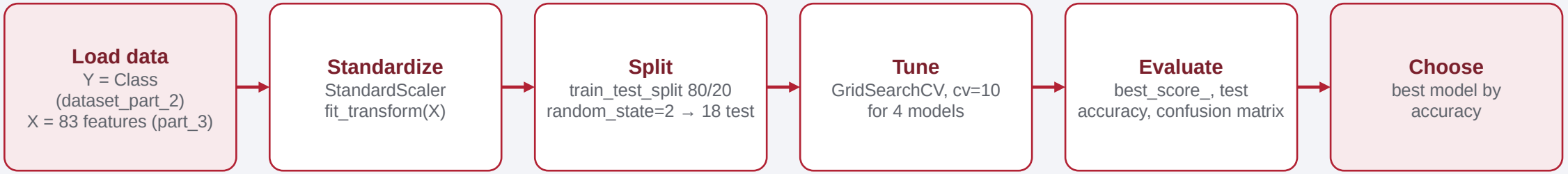
Component	Type	Why I added it
Launch site dropdown	Interaction	Switch between all sites and one site, so every chart can be compared site by site
Success pie chart	Plot	All sites: share of successful landings per site. One site: its success vs. failure split
Payload range slider	Interaction	0 to 10,000 kg in 1,000 kg steps, to zoom in on a payload range
Payload vs. outcome scatter	Plot	Each launch by payload and class, coloured by booster version, to show which payloads and boosters land

Two callbacks connect them: the dropdown updates the pie chart, and the dropdown plus slider together update the scatter plot.

The pie and scatter work as a pair: the pie answers "where", the scatter answers "which payload and booster".

GitHub app: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/spacex_dash_app.py

Predictive Analysis (Classification)



Model	Hyperparameters searched
Logistic Regression	C: 0.01, 0.1, 1 · penalty: l2 · solver: lbfgs
Support Vector Machine	kernel: linear, rbf, poly, sigmoid · C and gamma: 10^-3 to 10^3
Decision Tree	criterion: gini, entropy · splitter · max_depth 2–18 · max_features: sqrt, log2 · min_samples leaf/split
K-Nearest Neighbors	n_neighbors 1–10 · algorithm: auto, ball_tree, kd_tree, brute · p: 1, 2

Takeaway Each model was tuned on the training data only, then scored once on the 18 held-out launches.

GitHub notebook: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

Exploratory data analysis

- Success climbs with experience: 0% → 90%
- KSC LC-39A is the strongest site
- GTO is the hardest orbit (52%)

Interactive analytics

- Launch sites hug the coast, far from cities
- Payloads of 2,000–5,000 kg land most often
- FT and B4 boosters outperform v1.0/v1.1

Predictive analysis

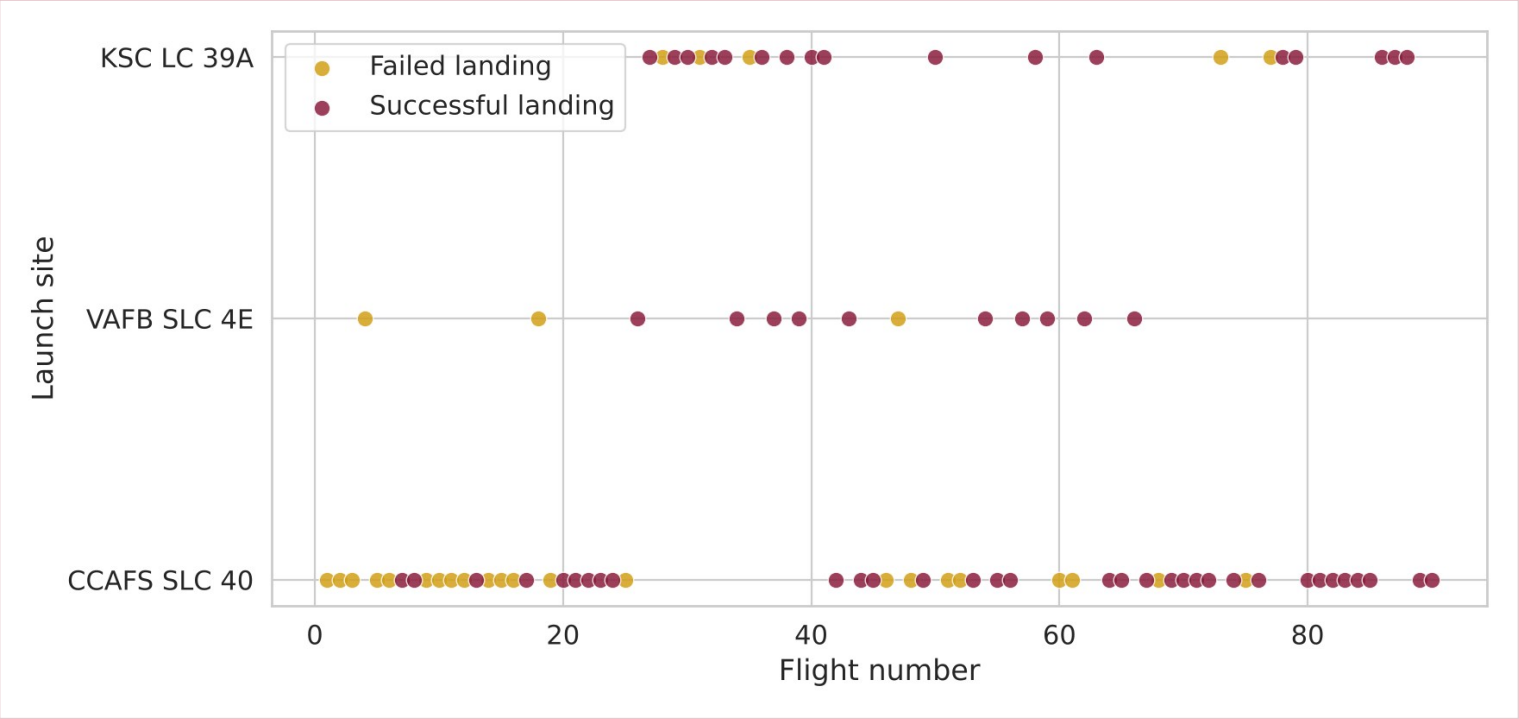
- 4 models, all 83.3% on test data
- Decision Tree has the best CV score
- Errors are false 'will land' predictions

Takeaway Landing success is predictable from launch features, and it has improved steadily as SpaceX gained experience.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

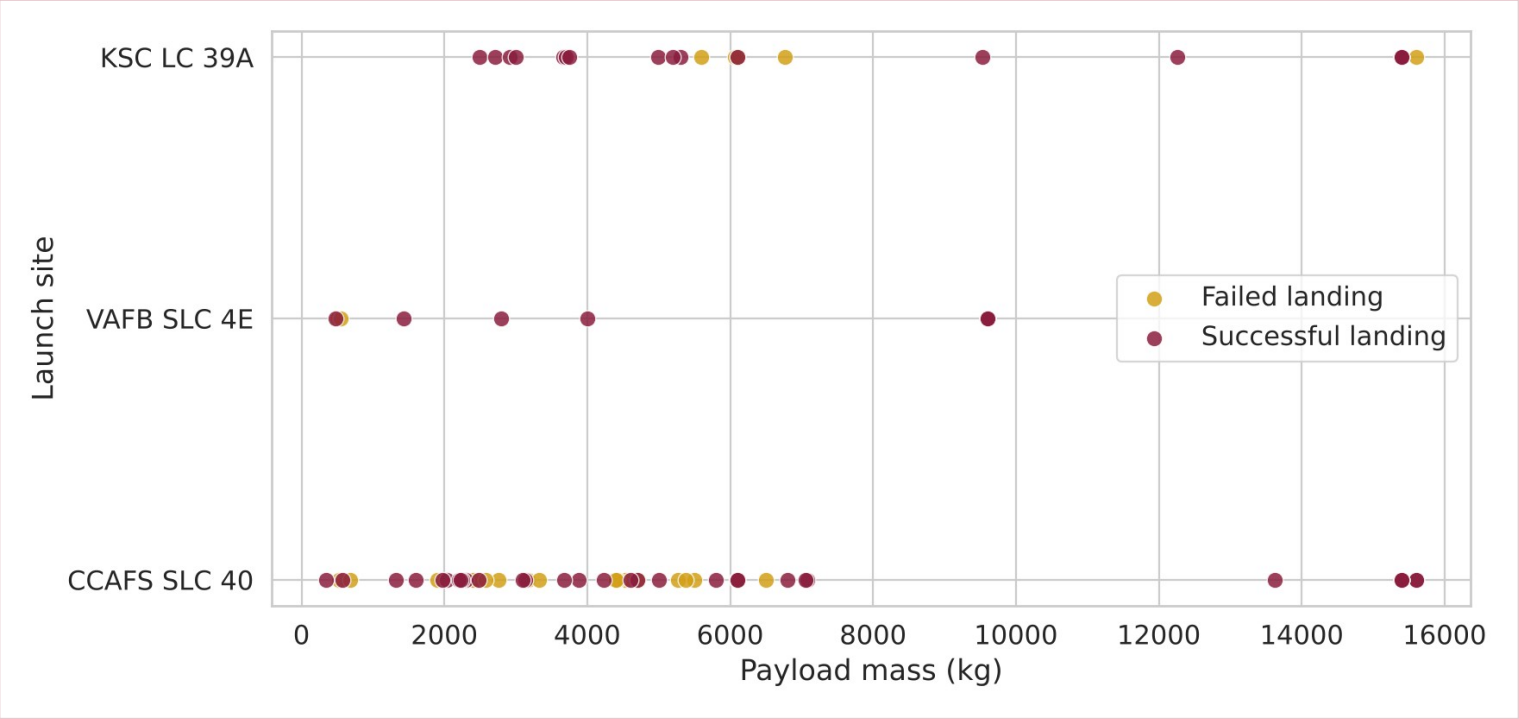


What it shows

- Early flights (mostly from CCAFS SLC 40) were mostly failures
- Landings become mostly successful after about flight 20 at every site
- CCAFS SLC 40 carries the most launches (55 of 90)

Takeaway More flights, more landings: experience is the strongest pattern in the data.

Payload vs. Launch Site

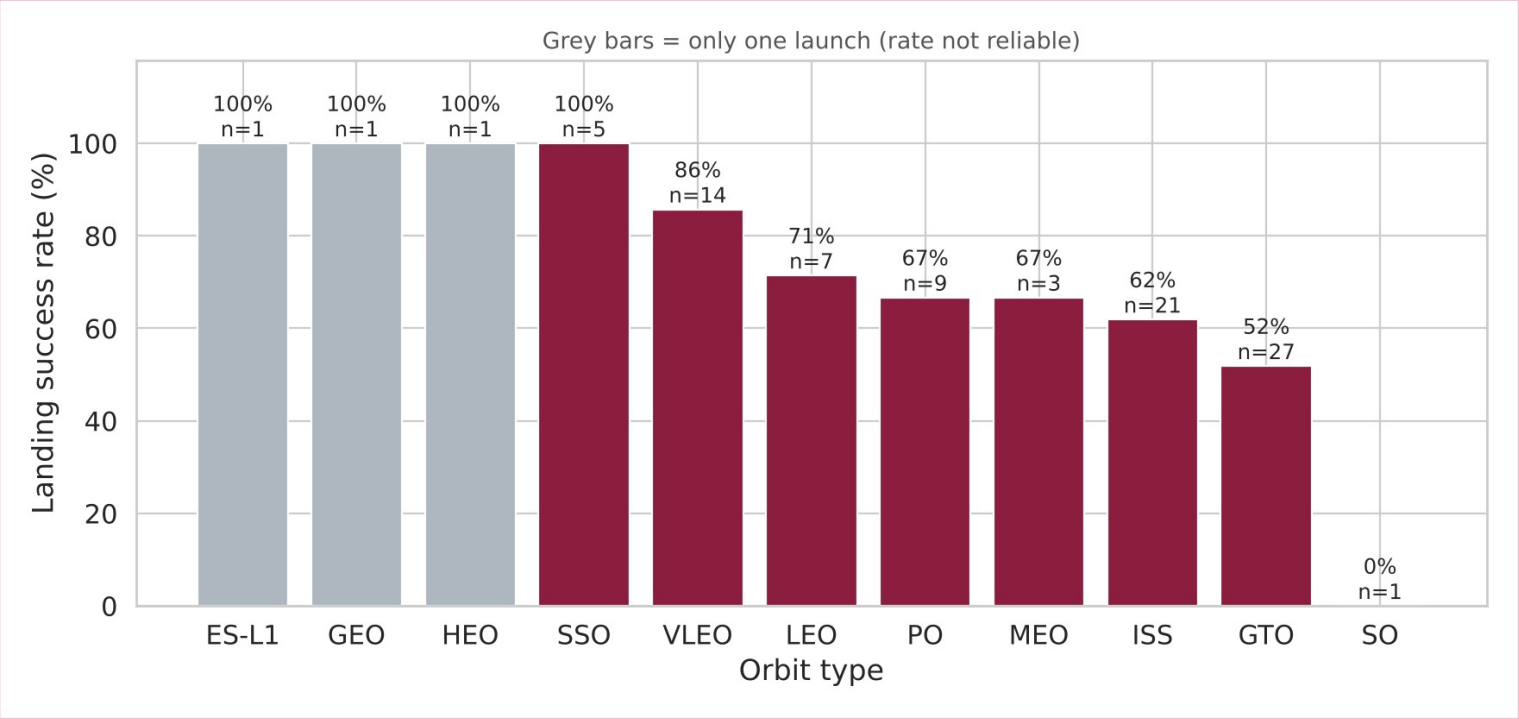


What it shows

- VAFB SLC 4E launched no payloads above 10,000 kg
- Heavy payloads (over 9,000 kg) mostly landed successfully
- Light and medium payloads show a mix of outcomes

Takeaway Heavier payloads did not hurt landing success here, partly because they came later in the program.

Success Rate vs. Orbit Type

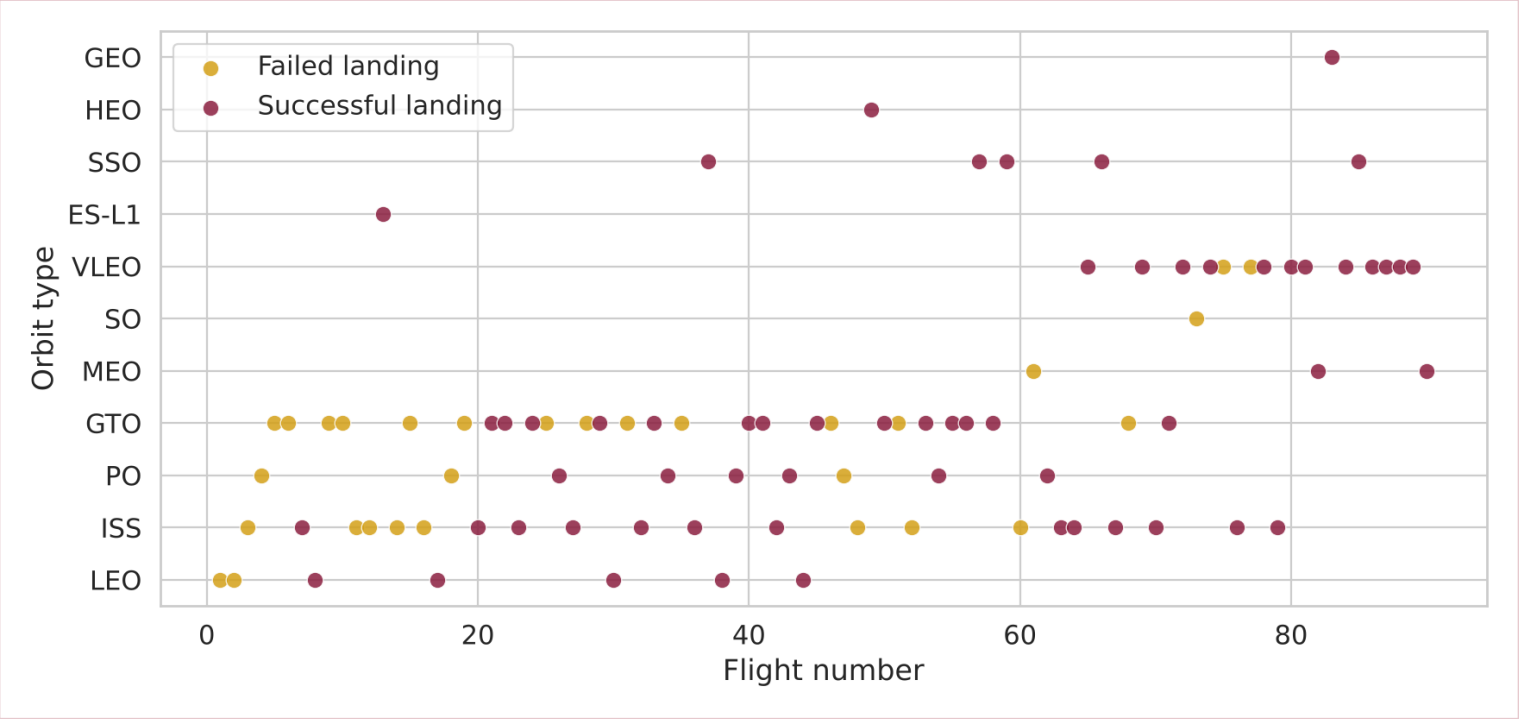


What it shows

- ES-L1, GEO, HEO and SSO show 100%, but the first three rest on one launch each (grey)
- SSO (5 of 5) and VLEO (86%) are reliably strong
- GTO, the most common orbit, has the lowest rate with real volume (52%)

Takeaway Read rates together with launch counts: a 100% from one launch is not evidence.

Flight Number vs. Orbit Type

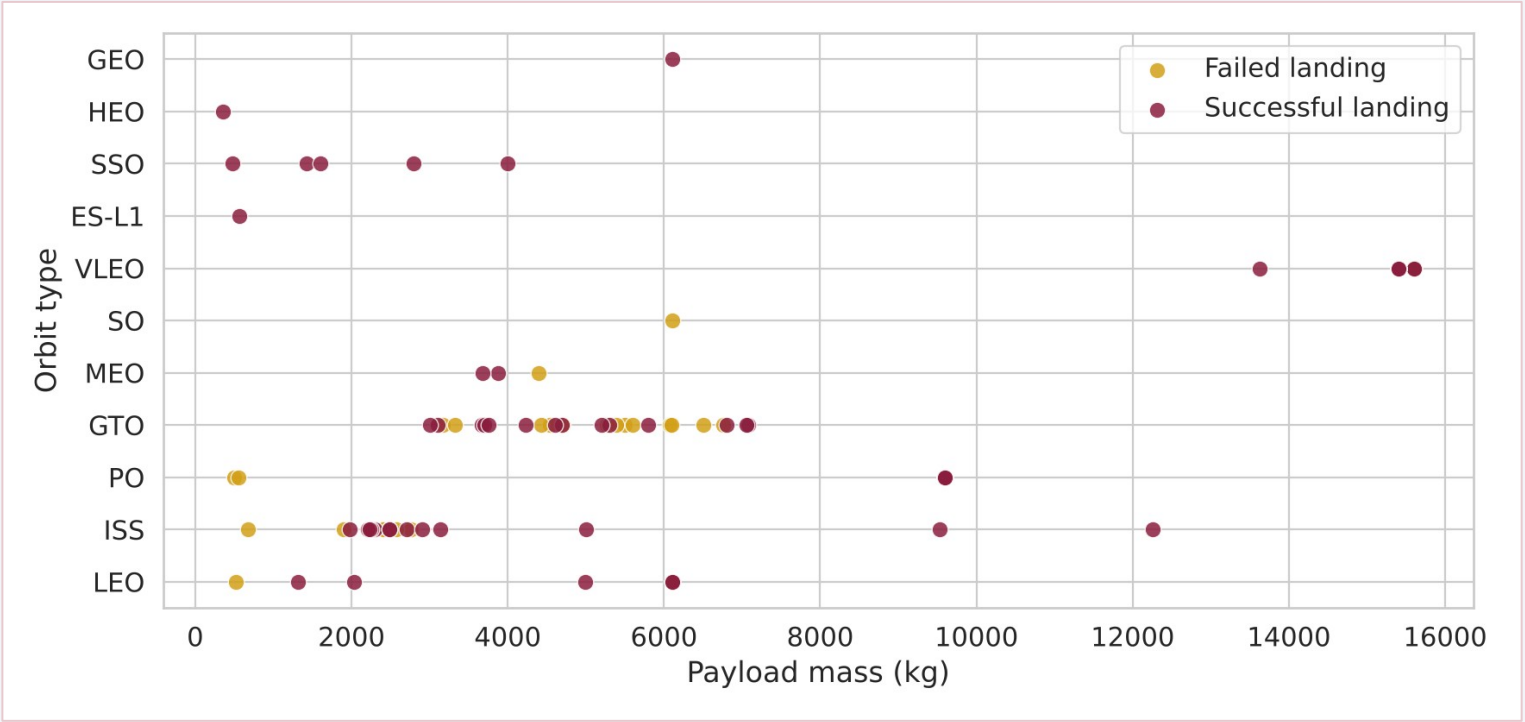


What it shows

- LEO success rises with flight number
- GTO shows no clear link to flight number
- VLEO appears only in later flights, and mostly lands

Takeaway Experience helps in most orbits, but GTO stays hard even for later flights.

Payload vs. Orbit Type

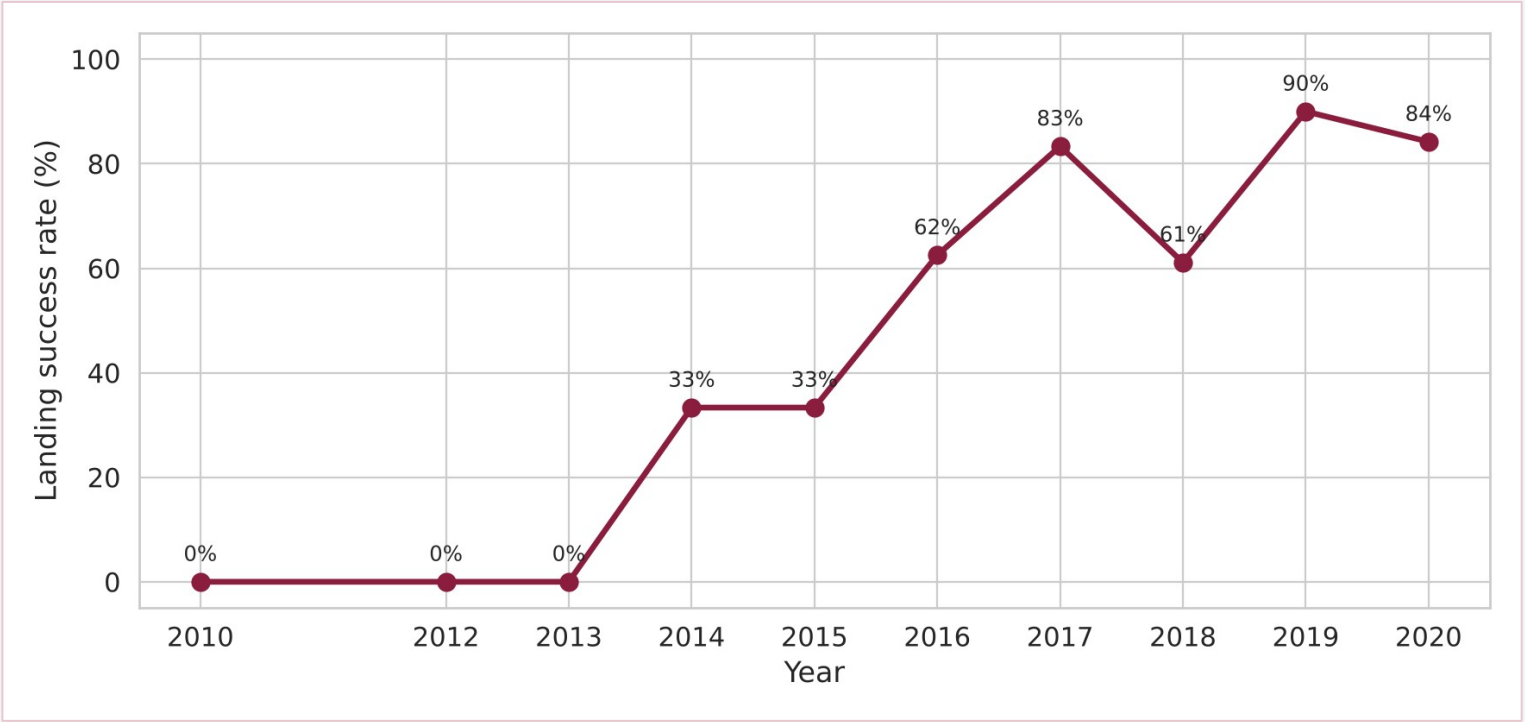


What it shows

- Heavy payloads land well for PO, LEO and ISS
- GTO successes and failures overlap across 3,000–7,000 kg
- VLEO missions carry 13,000–15,600 kg and mostly land

Takeaway Payload weight matters differently by orbit; for GTO it does not separate outcomes.

Launch Success Yearly Trend



What it shows

- No successful landings from 2010 to 2013
- Success rose steadily from 2013 to 2017
- A dip in 2018 (61%), then 90% in 2019 and 84% in 2020

Takeaway Landing success went from 0% to around 85–90% in seven years.

All Launch Site Names

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

4

launch sites

Takeaway DISTINCT returns each launch site once: four sites, three in Florida and one in California.

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster	Launch site	Payload	Mass (kg)	Orbit	Customer	Landing outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	LEO (ISS)	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	No attempt

Takeaway LIKE 'CCA%' matches every Cape Canaveral site; the earliest records are all v1.0 boosters with no landing.

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_kg
     FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

total_payload_kg
45,596

45,596 kg

total payload for NASA (CRS)

Takeaway Boosters launched for NASA's Commercial Resupply Services carried 45,596 kg in total.

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload_kg
     FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

avg_payload_kg
2,928.4

2,928.4 kg

average payload per F9 v1.1 launch

Takeaway The F9 v1.1 booster carried an average of 2,928.4 kg per launch.

First Successful Ground Landing Date

```
%sql SELECT MIN(Date) AS first_ground_pad_success
     FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

first_ground_pad_success
2015-12-22

Dec 22, 2015

first successful ground-pad landing

Takeaway The first successful ground-pad landing was on 22 December 2015, five years after the first launch.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE
     WHERE Landing_Outcome = 'Success (drone ship)'
       AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

4 FT

boosters: drone ship + 4,000–6,000 kg

Takeaway Four boosters landed on a drone ship with a 4,000–6,000 kg payload, all Full Thrust (FT) versions.

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) AS total
     FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

99%

mission success (100 of 101)

Takeaway 100 of 101 missions succeeded (the second 'Success' row is a label with a trailing space). Mission success is not the same as landing success.

Boosters Carried Maximum Payload

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE
     WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version	Booster_Version	Booster_Version
F9 B5 B1048.4	F9 B5 B1049.4	F9 B5 B1051.3
F9 B5 B1056.4	F9 B5 B1048.5	F9 B5 B1051.4
F9 B5 B1049.5	F9 B5 B1060.2	F9 B5 B1058.3
F9 B5 B1051.6	F9 B5 B1060.3	F9 B5 B1049.7

15,600 kg

maximum payload mass

Takeaway A subquery finds the maximum (15,600 kg); all 12 boosters that carried it are Block 5, the newest version.

2015 Launch Records

```
%sql SELECT substr(Date,6,2) AS month, Landing_Outcome, Booster_Version, Launch_Site
     FROM SPACEXTABLE
     WHERE substr(Date,0,5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Takeaway Both 2015 drone-ship failures were v1.1 boosters from CCAFS LC-40, early attempts before landings worked.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT Landing_Outcome, COUNT(*) AS count FROM SPACEXTABLE
     WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
     GROUP BY Landing_Outcome ORDER BY count DESC;
```

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

31

landing outcomes in this period

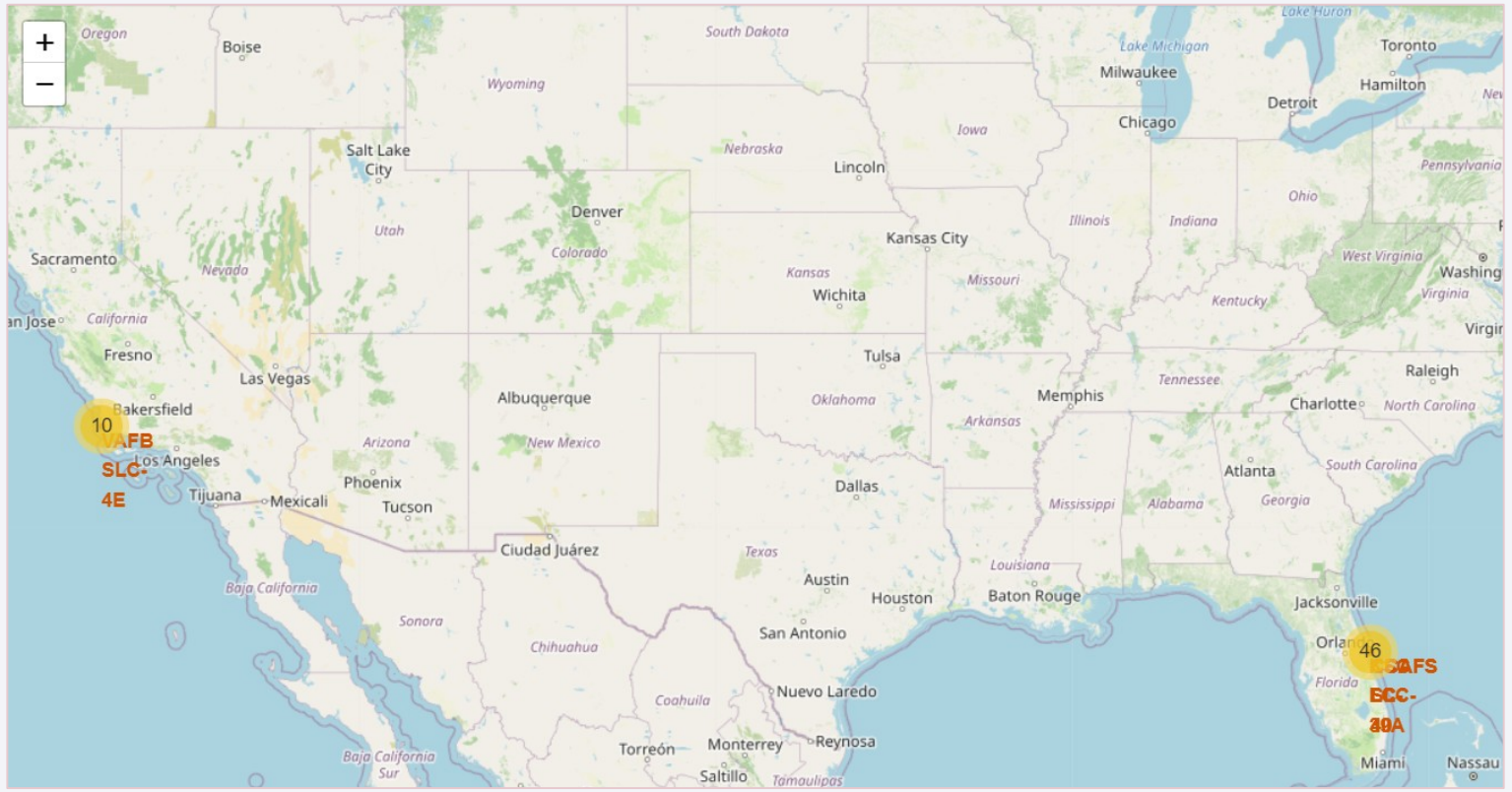
Takeaway In this early period 'No attempt' was most common, and drone-ship landings were 50/50 (5 successes, 5 failures).

Section 3

Launch Sites Proximities Analysis



All Launch Sites on the Map

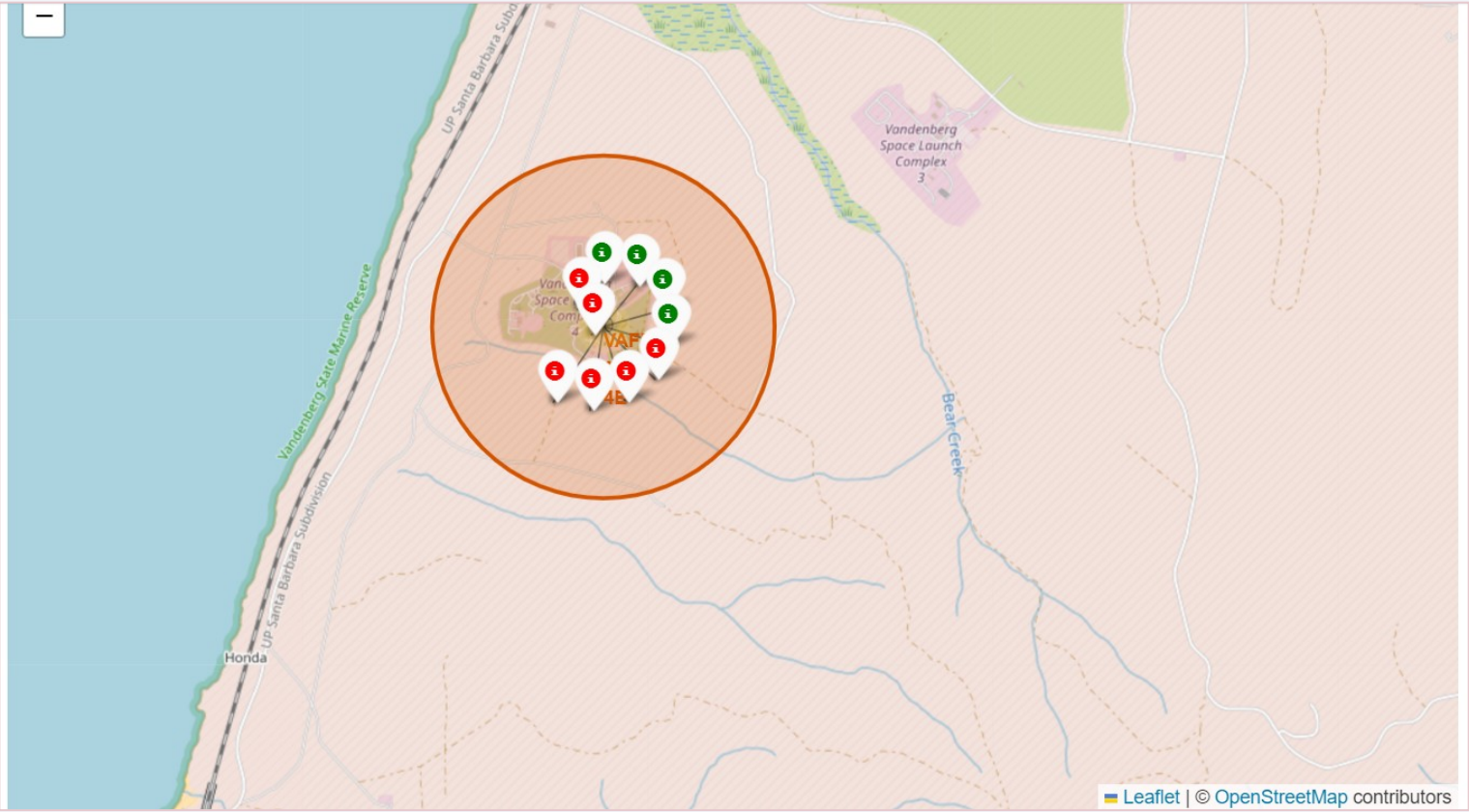


Key elements and findings

- Each launch site has a highlighted circle and a text label; clusters count the launches
- 1 site in California (VAFB SLC-4E, 10 launches) and 3 in Florida (46 launches)
- Every site sits right on the coast, and all are as far south as practical in the US mainland

Takeaway Sites are coastal and as far south as practical: rockets fly over water and gain speed from Earth's rotation.

Launch Outcomes at VAFB SLC-4E

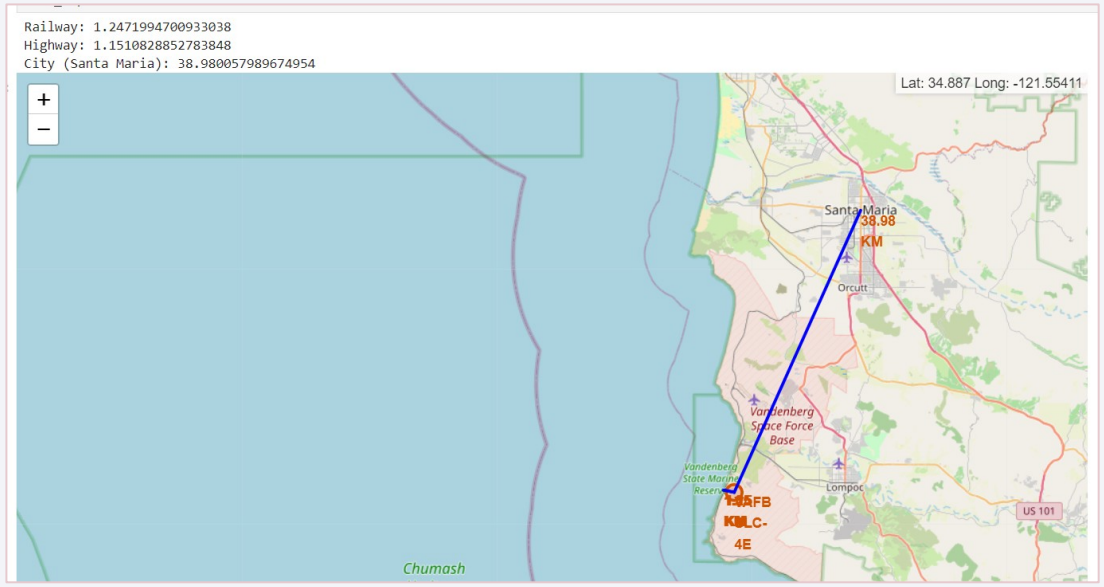


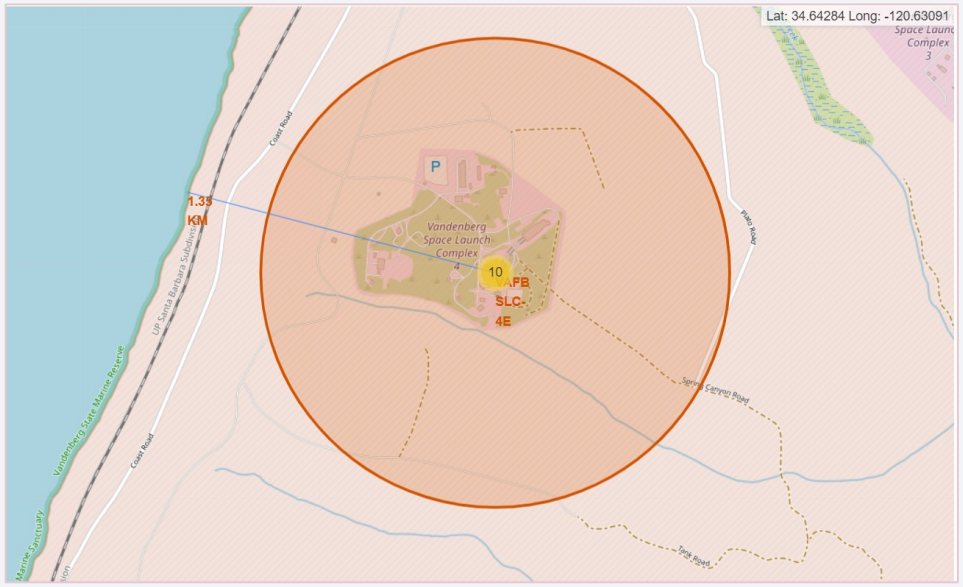
Key elements and findings

- Clicking a cluster spreads it into one marker per launch: green = landed, red = failed
- VAFB SLC-4E: 4 of 10 launches landed (40%)
- Across all sites, KSC LC-39A is best: 10 of 13 landed (77%)
- CCAFS LC-40 had the most launches (26) but only 7 landed

Takeaway Colour-coded markers show each site's success rate at a glance.

VAFB SLC-4E and Its Proximities

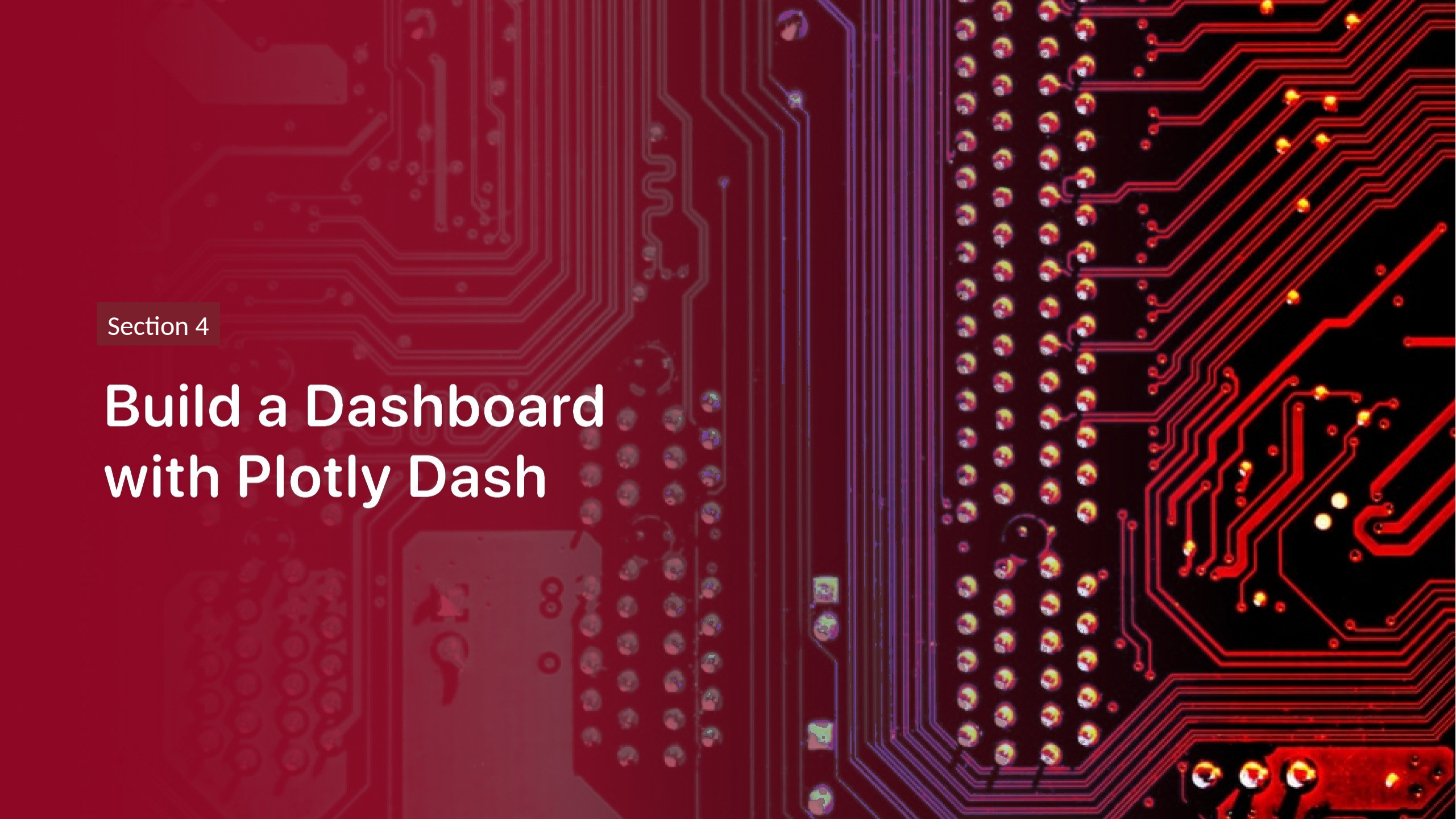




Key elements and findings (left: line to the nearest city; right: close-up of the line to the coast)

- Coastline 1.35 km away: rockets fly out over the Pacific
- Railway 1.25 km away: heavy parts and fuel can reach the pad
- Highway 1.15 km away: crews and equipment have road access
- Nearest city, Santa Maria, 38.98 km away: distance protects residents

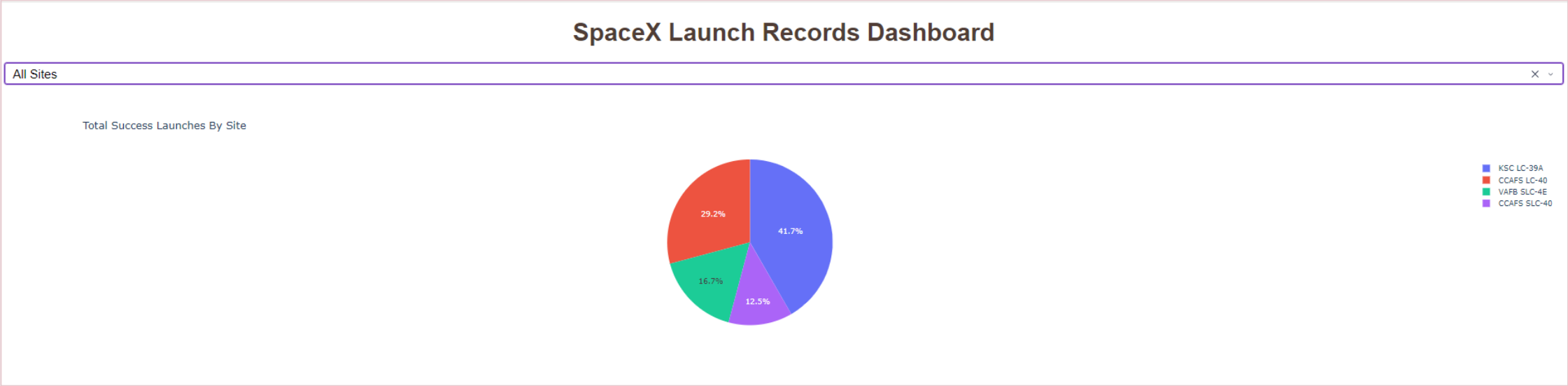
Takeaway Launch sites are close to transport and the coast, but kept far from cities.



Section 4

Build a Dashboard with Plotly Dash

Successful Launches by Site (All Sites)

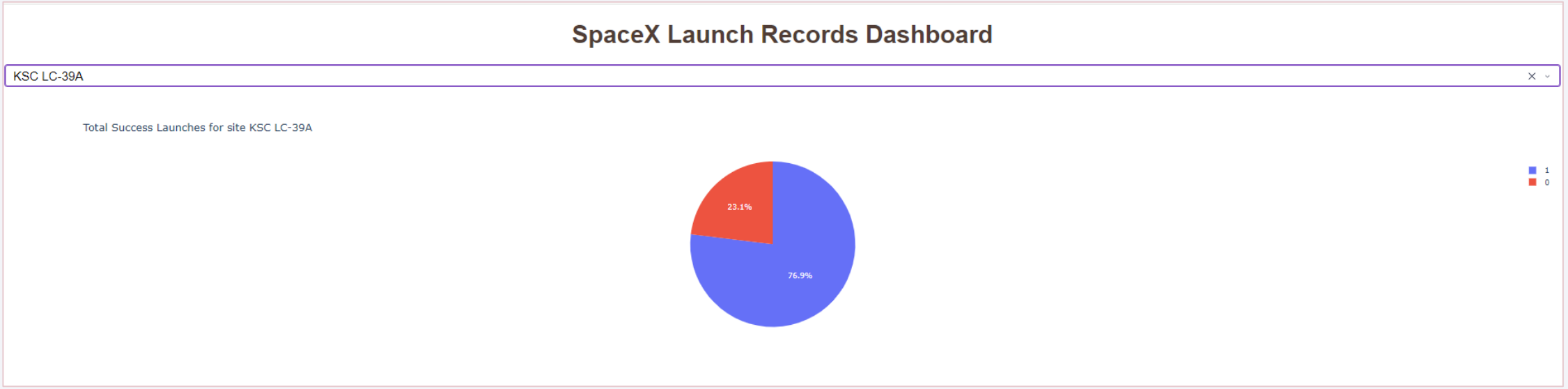


Key elements and findings

- Each slice is a site's share of all 24 successful landings
- KSC LC-39A: 41.7%
- CCAFS LC-40: 29.2%
- VAFB SLC-4E: 16.7%
- CCAFS SLC-40: 12.5%

Takeaway KSC LC-39A produced the most successful landings of any site.

KSC LC-39A: Highest Success Ratio



Key elements and findings

- Selected KSC LC-39A in the dropdown
- 76.9% of its launches landed (class 1)
- 23.1% failed (class 0): 3 of 13 launches

Takeaway KSC LC-39A has the best success ratio of all sites: about 3 in 4 launches land.

Payload vs. Outcome, All Payloads (0–10,000 kg)



Key elements and findings

- y = 1 is a landing, y = 0 is a failure; colour = booster version
- FT boosters have the most successes
- v1.0 and v1.1 boosters almost never landed (1 of 20)
- Few launches above 6,000 kg, with mixed results

Takeaway Booster version matters as much as payload: newer boosters land far more often.

Payload vs. Outcome, 3,000–7,000 kg Selected



Key elements and findings

- Range slider set to 3,000–7,000 kg
- 13 of 28 launches landed (46%)
- FT boosters: 8 of 14 landed; B4: 4 of 7
- v1.1 boosters: 0 of 6 in this range
- All 5 launches above 5,500 kg failed

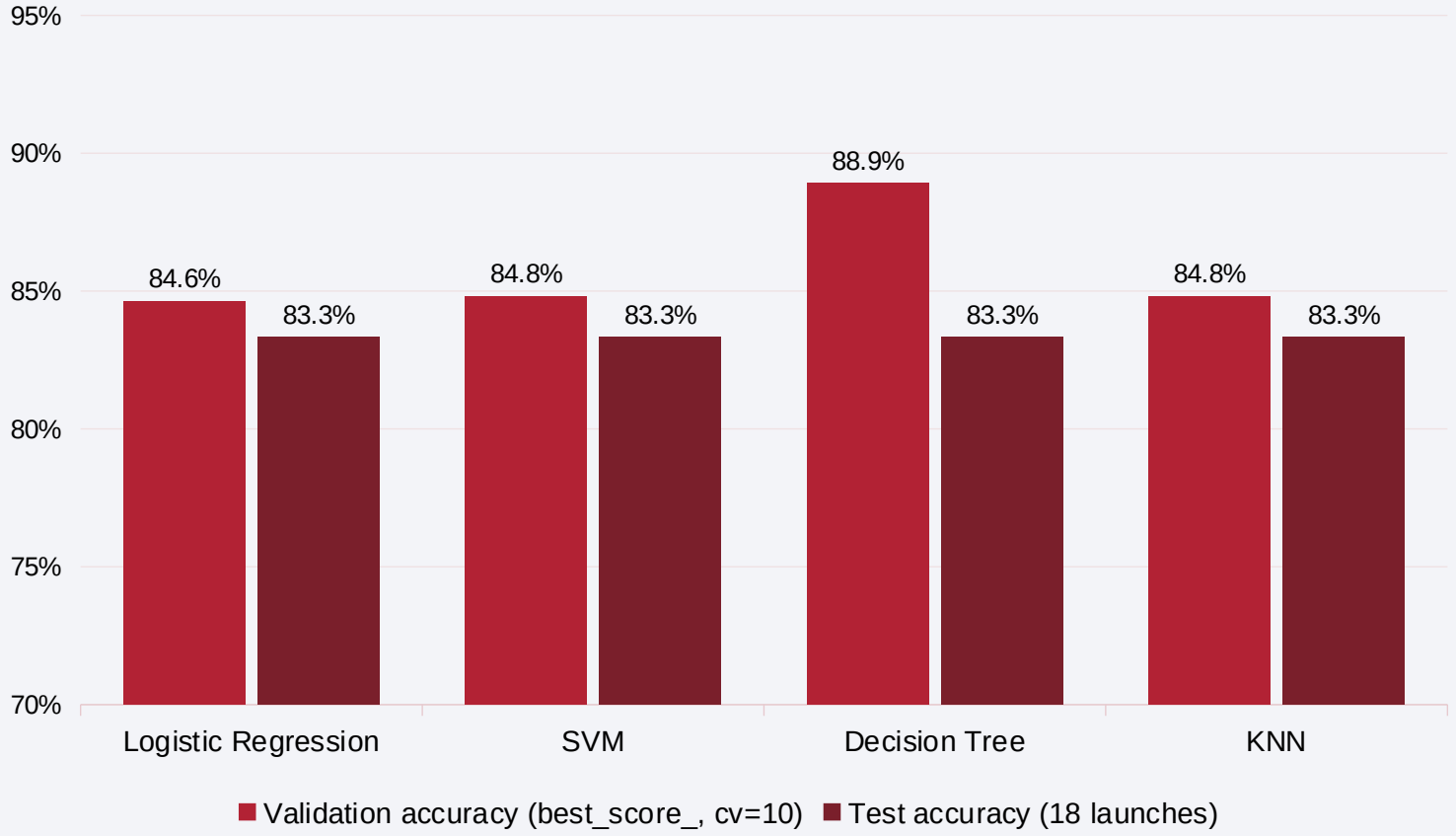
Takeaway Mid-weight payloads on FT or B4 boosters land most often; v1.1 boosters never did.

Section 5

Predictive Analysis (Classification)



Classification Accuracy

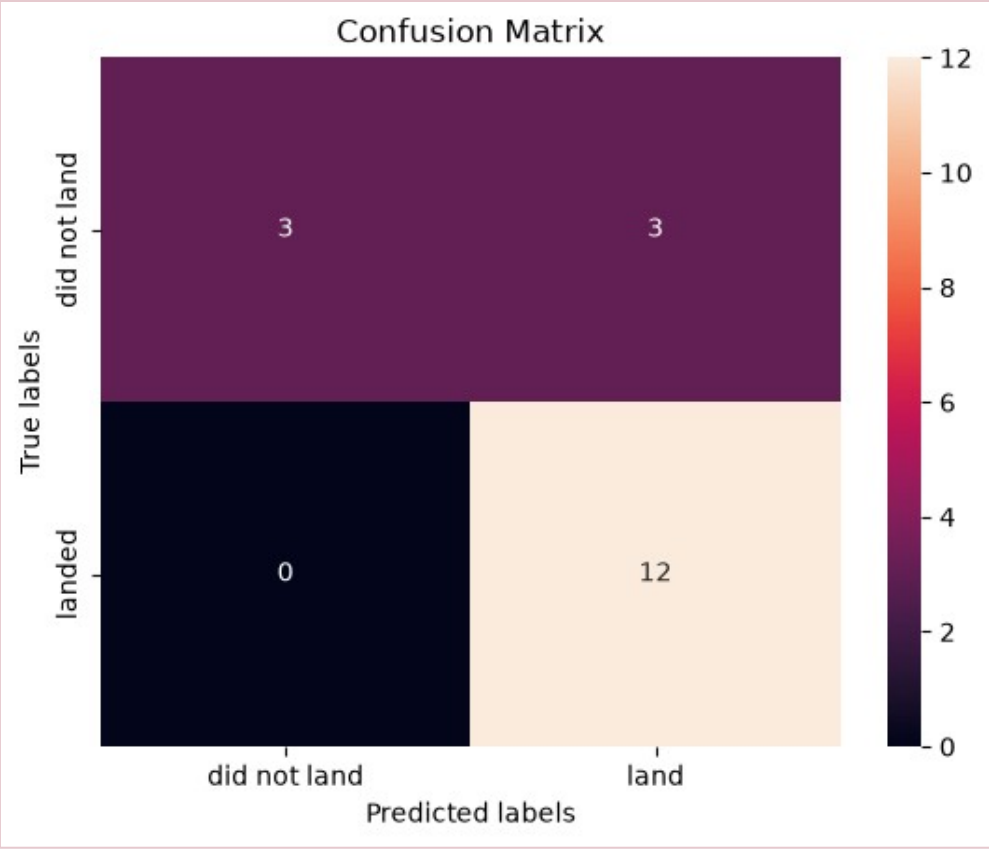


Which model is best?

- On the test set, all four models tie at 83.3% (15 of 18 correct)
- The Decision Tree has the highest validation score (88.9%), so it is selected as the best model
- Best tree: gini, max_depth 6, max_features sqrt, min_samples_leaf 4, min_samples_split 5
- With only 18 test launches, one mistake moves accuracy by 5.6 points, so the gap between models is small

Takeaway Best model: Decision Tree (highest cross-validation accuracy); all models reach 83.3% on unseen launches.

Confusion Matrix



Decision Tree on the test set (18 launches)

- **12 true positives:** predicted landed, and did land
- **3 true negatives:** predicted no landing, and none happened
- **3 false positives:** predicted a landing that failed
- **0 false negatives:** it never missed a real landing

The model is good at recognising landings but too optimistic about failures.

Takeaway False positives are the main problem: for a bidder, they mean underestimating SpaceX's true launch cost.

Conclusions

1. **Experience drives success:** Landing success rose from 0% (2010–13) to 90% (2019) as SpaceX flew more missions.

2. **Location matters:** KSC LC-39A has the best success ratio (77%); all sites are coastal, near transport, and far from cities.

3. **Payload and booster:** 2,000–5,000 kg payloads on FT or B4 boosters land about 80% of the time; v1.0/v1.1 boosters rarely landed.

4. **Orbit type:** SSO and VLEO land reliably; GTO, the most common orbit, is the hardest at 52%.

5. **Prediction works:** A tuned Decision Tree predicts landings with 83.3% test accuracy, giving Space Y a basis for estimating SpaceX's costs.

Innovative Insights: Beyond the Labs

Landing is a learned skill

- Grouping launches into blocks of 20 flights: 25% → 80% → 70% → 75% → 100% landed. Program maturity is the strongest signal, so a bid model should weight recent launches more heavily.

Beware of 100% rates

- ES-L1, GEO and HEO show 100% success, each from a single launch. I flagged these in grey on the orbit chart so they are not mistaken for proven results.

The test set is too small to pick a winner

- 18 test launches means each error moves accuracy by 5.6 points, which is why all four models tie at 83.3%. Repeated cross-validation or more launch data would separate them.

Translate errors into dollars

- A false positive assumes a reusable booster, so SpaceX looks cheaper than it really is. Space Y should tune the decision threshold to cut false positives before using predictions in a bid.

Takeaway Resilient pipeline: when the live SpaceX API failed, the project switched to IBM's saved copy without changing any analysis code.

Appendix

Asset	File in the GitHub repository
SpaceX API data collection	`jupyter-labs-spacex-data-collection-api.ipynb`
Web scraping	`jupyter-labs-webscraping.ipynb`
Data wrangling	`labs-jupyter-spacex-Data_wrangling-v2.ipynb`
EDA with SQL	`jupyter-labs-eda-sql-coursera_sqllite.ipynb`
EDA with data visualization	`jupyter-labs-eda-dataviz-v2.ipynb`
Folium launch-site maps	`lab-jupyter-launch-site-location-v2.ipynb`
Plotly Dash dashboard	`spacex_dash_app.py`
Machine learning prediction	`SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb`

```
# Landing label (data wrangling)
bad_outcomes = set(landing_outcomes.keys()
[[1,3,5,6,7]])
landing_class = [0 if o in bad_outcomes else 1
                 for o in df['Outcome']]
df['Class'] = landing_class

# Model tuning (predictive analysis)
tree_cv = GridSearchCV(tree, parameters, cv=10)
tree_cv.fit(X_train, Y_train)
tree_cv.score(X_test, Y_test)
```

Repository: https://github.com/RoShawn-F/spacex-falcon9-landing-prediction

Data: IBM Skills Network copies of the SpaceX API response and Wikipedia launch table (Falcon 9 launches, June 2010 – November 2020).

Thank you!

